
ASSIGNMENT

COMP50008

Enterprise Cloud and Infrastructure Automation 1 COM24A1-COMCT24A1

HAND OUT DATE: 24-Nov-25

HAND IN DATE: 06-Feb-26

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INSTRUCTION TO CANDIDATES:

1. Students are advised to underpin their answers with the use of references (cited using the Harvard Name System of Referencing).
2. Late submission will be awarded zero (0) unless Exceptional Circumstances (EC) are upheld.
3. We draw your attention to the [Academic Conduct](#) Procedure to encourage good academic practice.
4. Assignment presentation and modeling work should be submitted as a softcopy

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1. Introduction

FreshMart is a small, to, medium, sized organic food product business in Sri Lanka that is currently in the growth phase. At present, the business is mainly running through manual processes and face, to, face sales, which results in limited scalability and customer reach. Due to rising competition and customer demand, FreshMart is in need of a state, of, the, art, cloud, based digital infrastructure.

This report recommends an Amazon Web Services (AWS) cloud solution that will enable FreshMart's digital transformation. The proposed solution boasts of scalability, availability, security, performance, and cost efficiency, alongside compliance with data protection regulations.

2. Business Requirements Analysis

2.1 Business Requirements

- An online platform to showcase and sell products
- High availability to ensure continuous service
- Scalability to handle increased customer demand
- Secure storage of business and customer data
- Cost-effective infrastructure suitable for an SME
- Compliance with data security and privacy regulations

2.2 Technical Requirements

- Web hosting for business website
- Static content hosting (images, product pages)
- Database for customer and product information
- Secure user and admin access
- Network isolation for security
- Backup and disaster recovery capability

2.3 Mapping Business Needs to AWS Services

Business Need	AWS Service	Why
Web hosting	EC2, S3, CloudFront	Reliable, scalable, global reach
Database for products/orders	RDS (PostgreSQL) or DynamoDB	Managed, secure, compliant
Security & access control	IAM, KMS, AWS Shield	Data encryption, user roles, DDoS protection
Content delivery & low latency	CloudFront + Route 53	Fast loading for Sri Lankan customers
Serverless backend for orders	Lambda	Cost-efficient, auto-scaling
Backup & compliance	S3 versioning + AWS Backup	Meets data retention policies

3. Proposed Cloud Architecture

3.1 Overview of Architecture

The recommended cloud solution is an Amazon Virtual Private Cloud (VPC) set up to deliver a secure and secluded network environment. The design stabilizes a three, tier concept, isolating the presentation layer, the application layer, and the data layer. Using the layers, the system is more secure, easier to maintain, and scalable in the future.

Users access the FreshMart online platform through the internet. Their requests are first routed using Amazon Route 53, and to make it faster, Amazon CloudFront could be utilized. The static content is taken care of by Amazon S3, and the dynamic requests are processed by an Amazon EC2 instance that lives inside a public subnet. The data storage systems such as databases are hosted in private subnets to keep the data safe.

3. 2 Architecture Components

The network layer forms the foundation of the proposed architecture.

Virtual Private Cloud (VPC)

The cloud resources of FreshMart are isolated by setting up a customized VPC with a private IPv4 address range (CIDR block 10. 0. 0. 0/16). Centralized network governance becomes possible here, and by segregating FreshMarts infrastructure from other AWS tenants, security is boosted. (AWS, n.d.)

Public Subnet

A public subnet (10.0.1.0/24) is created within the VPC to host internet-facing resources. This subnet is associated with a route table that directs outbound traffic to the internet via an Internet Gateway

Internet Gateway & Routing

An Internet Gateway (IGW) is coupled with the VPC, enabling resources on a public subnet to access the internet. The route tables are arranged so that only public subnet traffic goes outside while private subnet traffic stays inside.

3.3 Compute Layer

Amazon EC2

Amazon Elastic Compute Cloud (EC2) is the platform for FreshMarts dynamic web application.

The web server software called Apache is running on the EC2 virtual machine along with other user interaction processes such as product searching and request filling.

EC2 gives a wide range of instance types catering to various performance and price levels. Moreover, the whole plan spots the way towards future upgrades such as Auto Scaling and Load Balancing to handle increased traffic. (AWS, 2026)

AWS Lambda

AWS Lambda is capable of powering backend activities like order processing, or notification handling. Such a serverless mode greatly cuts down the operational overhead and leads to better cost efficiency.

3.4 Storage Layer

Amazon S3

Amazon Simple Storage Service (S3) is chosen for hosting static website content like HTML files, images, and style sheets. Besides that, it is also being employed for data backups and logs.

S3 has great durability, availability, and scalability features which makes it a perfect choice for static website hosting and long, term data storage. (AWS, 2026)

Amazon EBS

Amazon Elastic Block Store (EBS) is attached to EC2 instances to store operating system files and application data. EBS provides persistent storage with snapshot features can be used for backup and recovery.

3.5 Database Layer

Amazon RDS

Amazon Relational Database Service (RDS) is a database platform where structured data such as product details, customer information, and records of transactions RDS is a fully managed database solution providing automated backups, patching, and monitoring. Setting RDS in a private subnet significantly increases the security level by blocking direct access from the public.

Amazon DynamoDB

For ultra, fast, scalable NoSQL operations, DynamoDB is a great option for storing session data or user activity logs.

3.6 Security Architecture

Security is a top priority and a key consideration in the design of all layers of the system architecture.

IAM

AWS Identity and Access Management (IAM) is a service that allows managing the users, roles, and permissions in an effective way. The use of role, based access control is a good practice to ensure that users are only given access to the AWS resources necessary for their jobs.

KMS

AWS Key Management Service (KMS) is a service used for encrypting data at rest in S3 and RDS, thus guaranteeing the confidentiality of the data and supporting compliance with data protection standards.

AWS Shield

AWS Shield is a Distributed Denial of Service (DDoS) protection service that keeps FreshMarts web application safe from malicious traffic. (AWS, 2026)

3.7 Performance & Content Delivery

Amazon CloudFront

Amazon CloudFront helps websites load faster by storing copies of the content in different edge locations around the world. So, the request does not have to travel far all the way to the origin server, this effectively reduces latency and improves user experience, particularly for customers who are geographically far from the server.

Amazon Route 53

Amazon Route 53 is a DNS service that it is robust and scalable. It is the one that directs the user traffic in an efficient way and guarantees the website is highly available.

3.8 Architecture Diagram

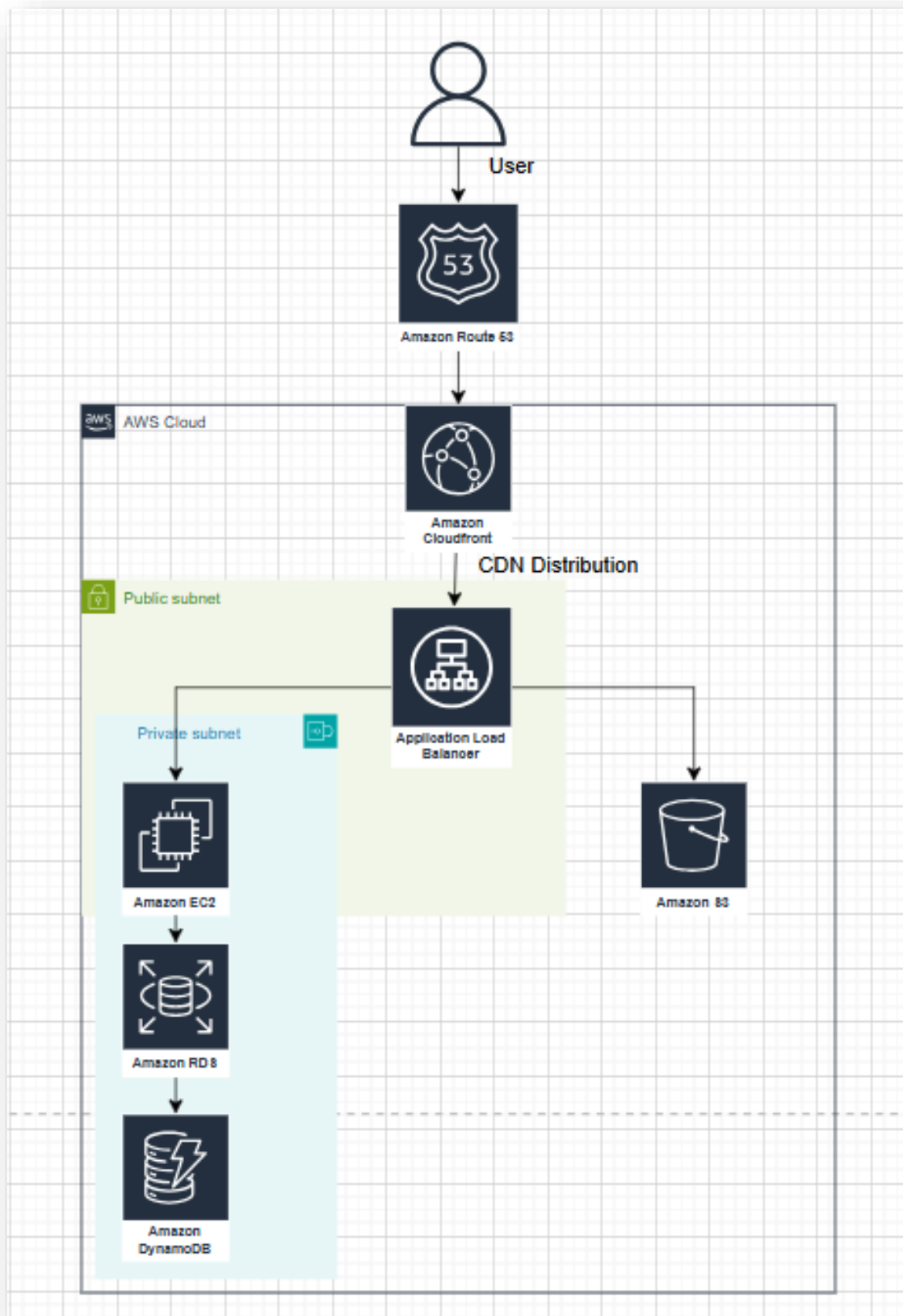


Figure 1 Architecture Diagram

4. Justification of Cloud Design

The AWS cloud architecture that is suggested for FreshMart aligns the company's challenges now and the anticipated expansion later.

Moreover, the chosen AWS functionalities have been carefully picked to enable high availability, along with scalability, security, cost, efficiency, and performance optimization, all vital for a growing SME in a competitive market.

4.1 High Availability

It is of utmost importance to have high availability for FreshMart to provide their customers with continuous access to their digital services.

AWS infrastructure uses highly available managed services capable of minimizing downtime and supporting system availability.

Amazon EC2 instances can be distributed across different availability zones (AZs), thus the system will be operable in case of failure of one of the zones. Amazon S3 ensures redundancy of data stored in different locations so even if one facility goes down, the data will still be accessible. In addition, Amazon RDS enables Multi, AZ installations, thus there will be automatic data replication to standby instances and a very short failover time in case of an incident. (Amazon, 2026)

Therefore, FreshMarts site will be available to the users continuously, which will lead to customer loyalty and uninterrupted running of the business.

4.2 Scalability

FreshMart wants to be able to ramp up their operations pronto in response to an increase in the customer base or seasonal sales. The cloud setup can scale up operations running on one or multiple servers.

Amazon EC2 allows for horizontal scaling through the creation of Auto Scaling Groups. This will enable the system to automatically launch new instances during times of peak usage and terminate them during off, peak periods. Amazon S3 and DynamoDB are

designed to be automatically scalable and without any need for a human operator, so they are perfectly suited for handling varying workloads.

This elastic nature of AWS ensures that FreshMart only pays for the resources it uses while maintaining optimal performance under varying demand levels.

4.3 Security and Compliance

Security is paramount as customer and business data are being processed. The cloud architecture suggested is in line with AWS security best practices and the shared responsibility model.

Strict access controls are implemented with AWS Identity and Access Management (IAM) using the least privilege principle. Sensitive backend components such as databases are placed in private subnets, thus no direct internet access is allowed. AWS Key Management Service (KMS) is utilized to encrypt data at rest, thus ensuring data privacy.

In addition, AWS Shield offers protection against Distributed Denial of Service (DDoS) attacks. Security groups and network access control lists (NACLs) help to limit inbound and outbound traffic. Such controls assist FreshMart in meeting data protection and regulatory requirements.

4.4 Cost Efficiency

Cost, efficiency is often the main worry for small and medium, sized enterprises. The use of AWS offers a pay, as, you, go pricing model, thus there is no need for upfront capital investment in hardware.

Amazon S3 and AWS Lambda are examples of services that reduce the operational expenses by eliminating server management. The option to pick right, sized EC2 instances is a great way for FreshMart to be sure that they do not over, provision. Auto Scaling, thus, adjusts resource usage according to demand, which is an additional cost optimizer.

This financially sound strategy enables FreshMart to dedicate their monetary resources towards expanding the business, rather than maintaining the infrastructure.

4.5 Performance Optimization

Performance has been optimized by leveraging AWS managed services that are inherently designed for low latency and high throughput. Content is delivered by Amazon CloudFront via globally distributed edge locations, which drastically shortens the time it takes for a website to load from a user's perspective.

High, performance computing is supported by optimized EC2 instance types, and Amazon RDS takes database performance to the next level through automated tuning and monitoring. The server load is lessened and the user experience is enriched by caching popular content.

These performance improvements result in rapid response times, better customer satisfaction, and a strong digital presence that FreshMart can leverage.

5. Practical Implementation – AWS

5.1 Task 1: VPC & EC2 Web Server

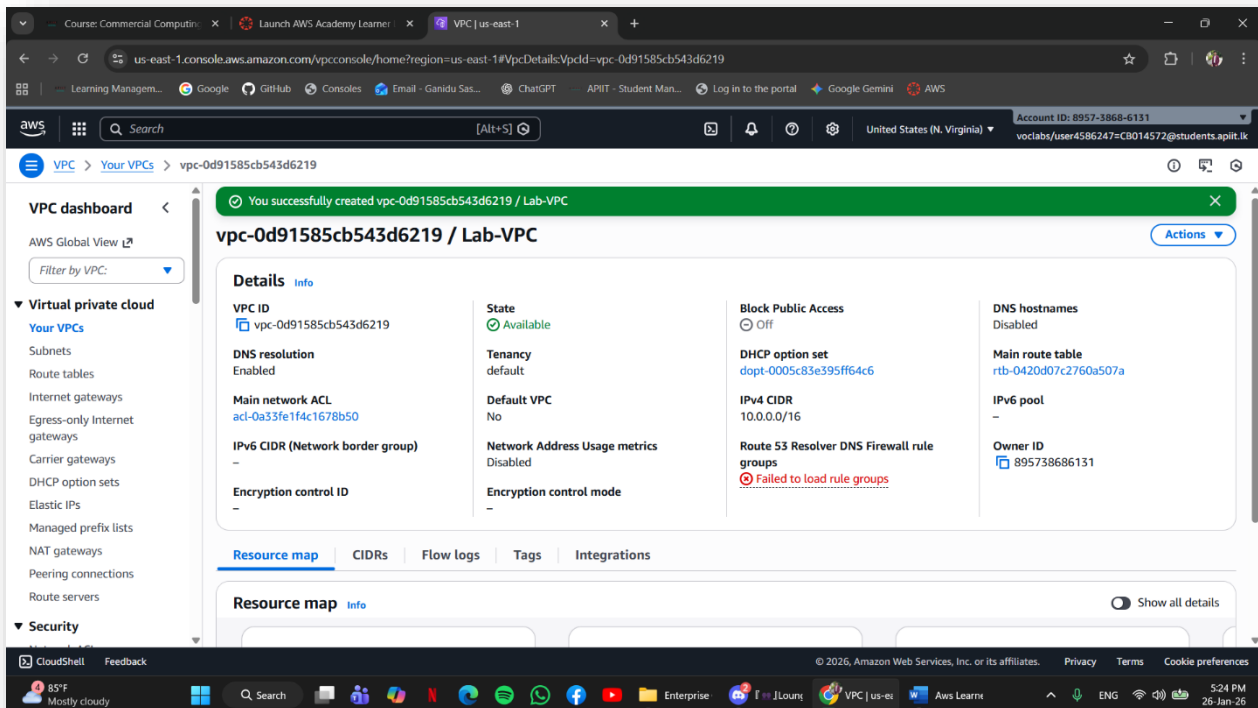


Figure 2 VPC Created

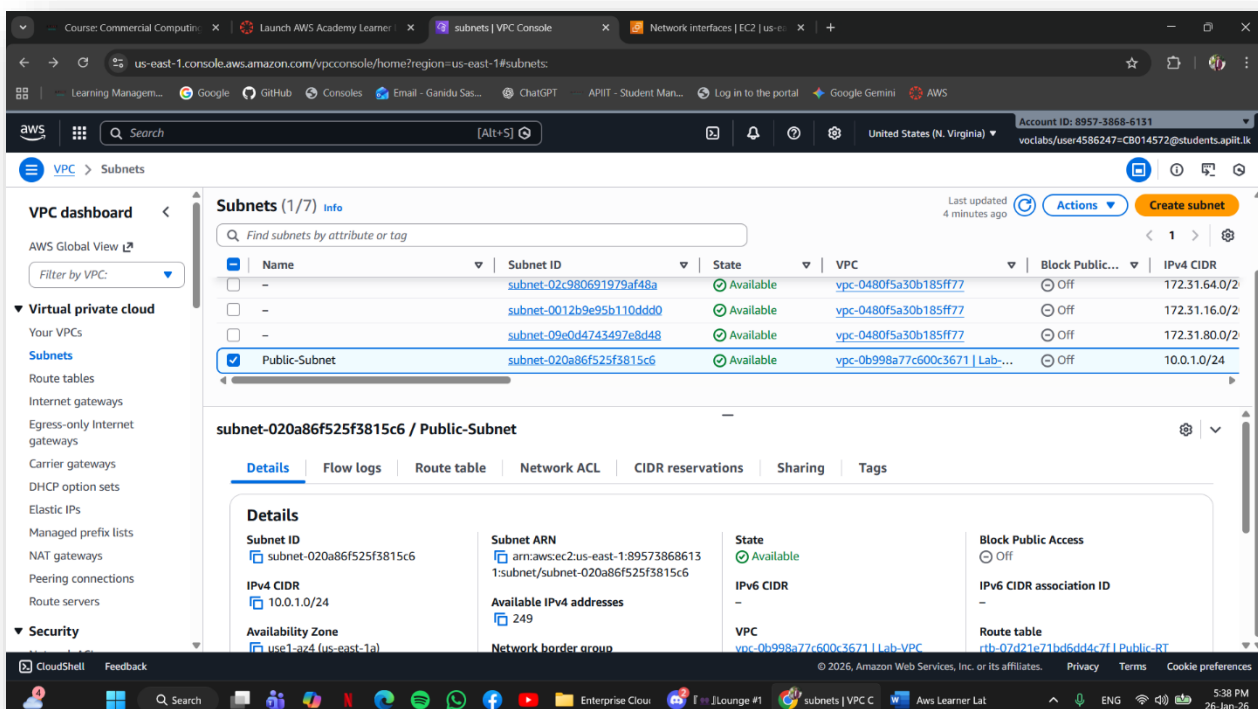


Figure 3 Public Subnet

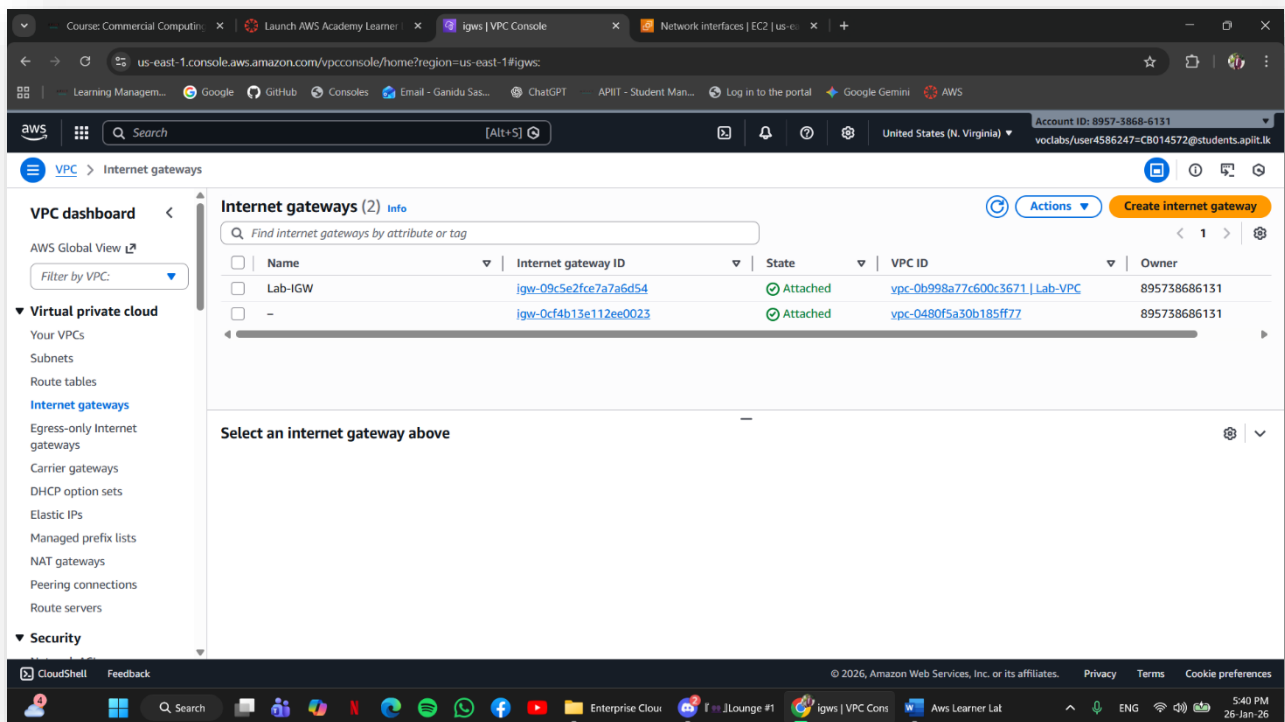


Figure 4 Internet Gateway Attached

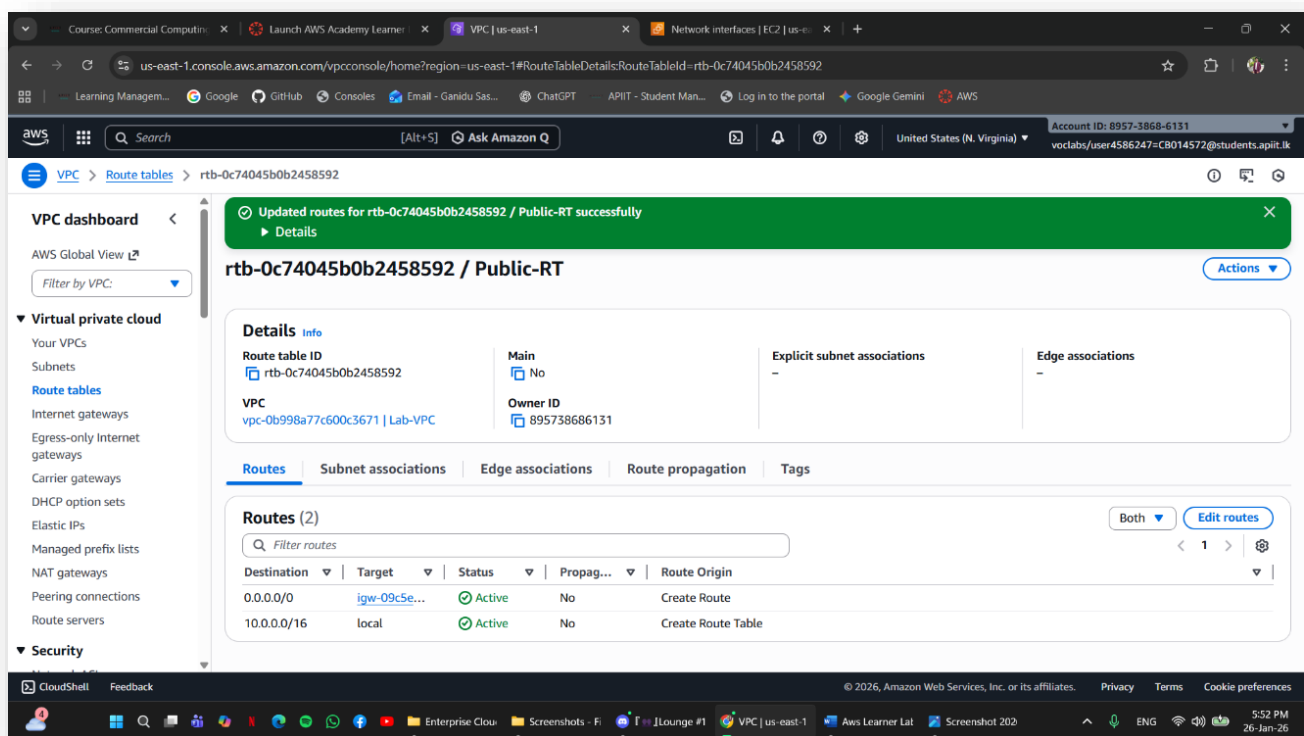


Figure 5 Public Route Table (0.0.0.0/0 → IGW)

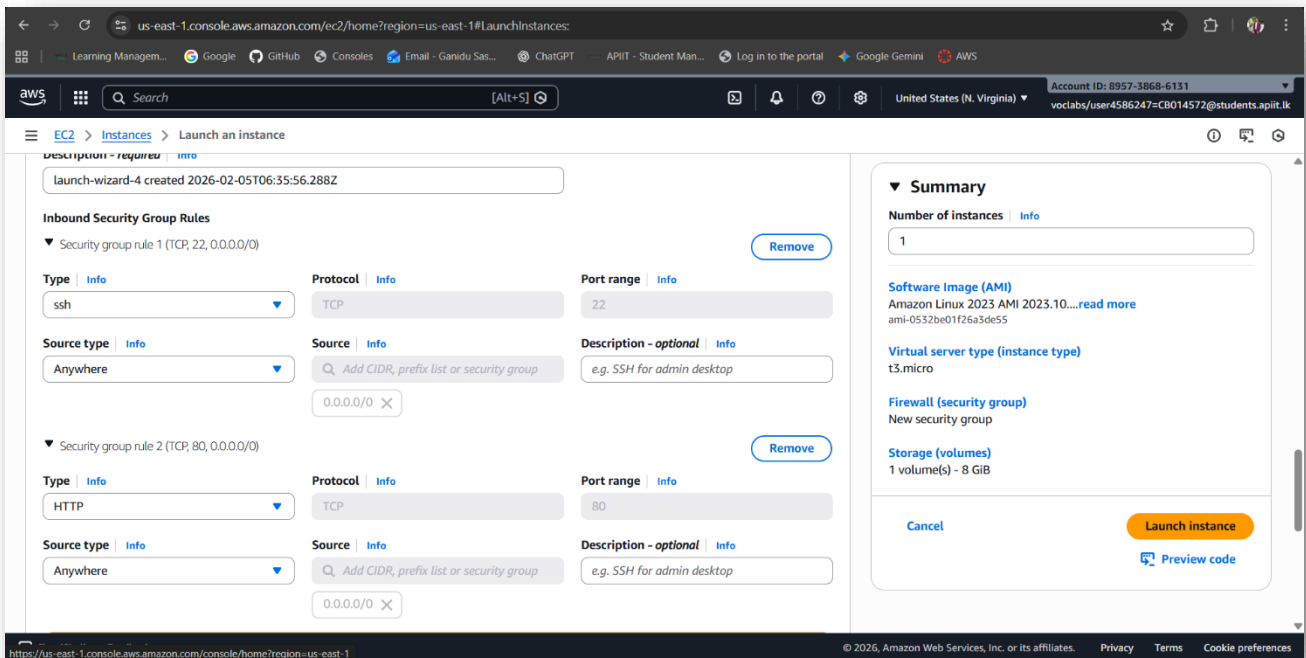


Figure 6 Security Group Rules (ADD)

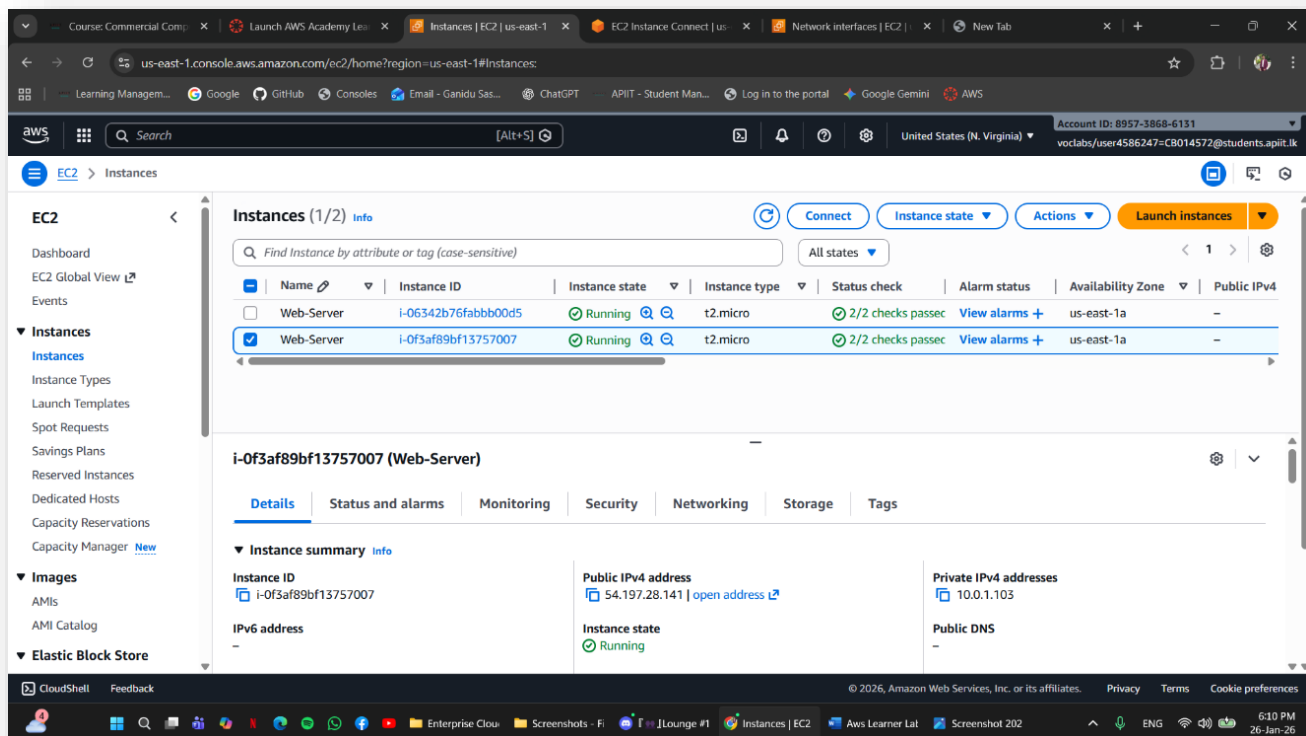


Figure 7 EC2 Instance Running

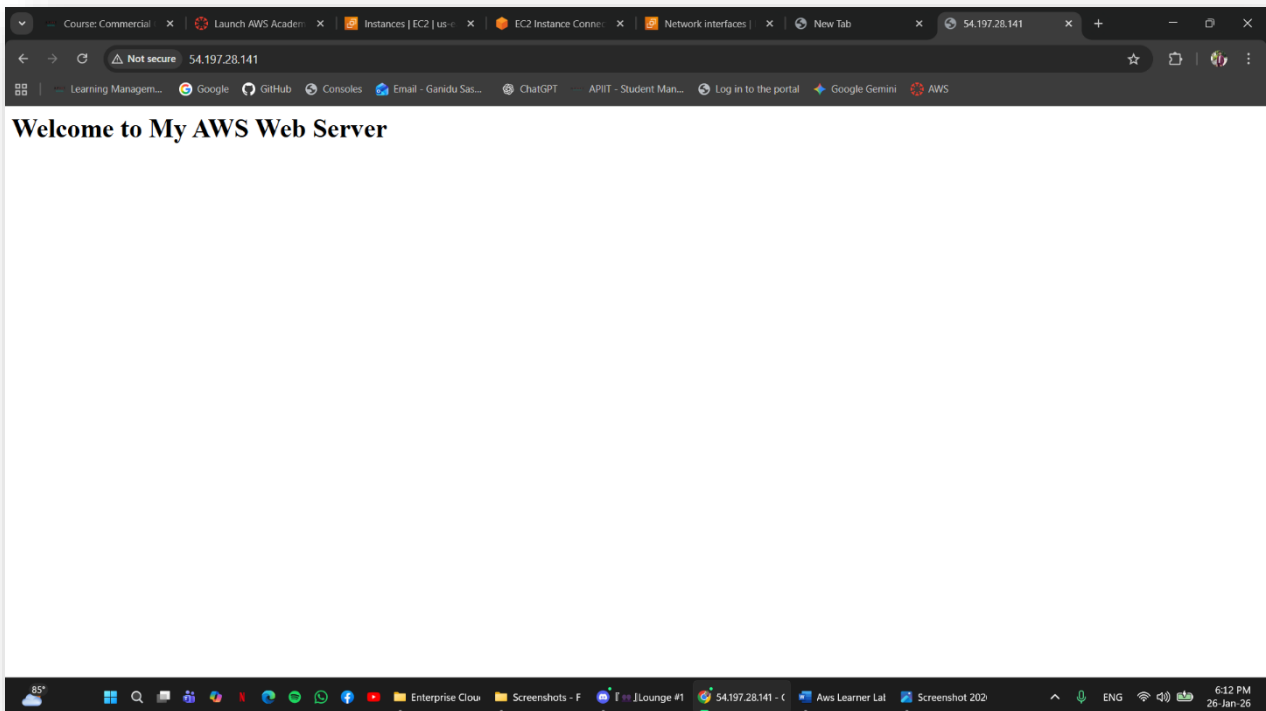


Figure 8 Browser showing EC2 Web Page

5.2 Task 2: S3 Static Website

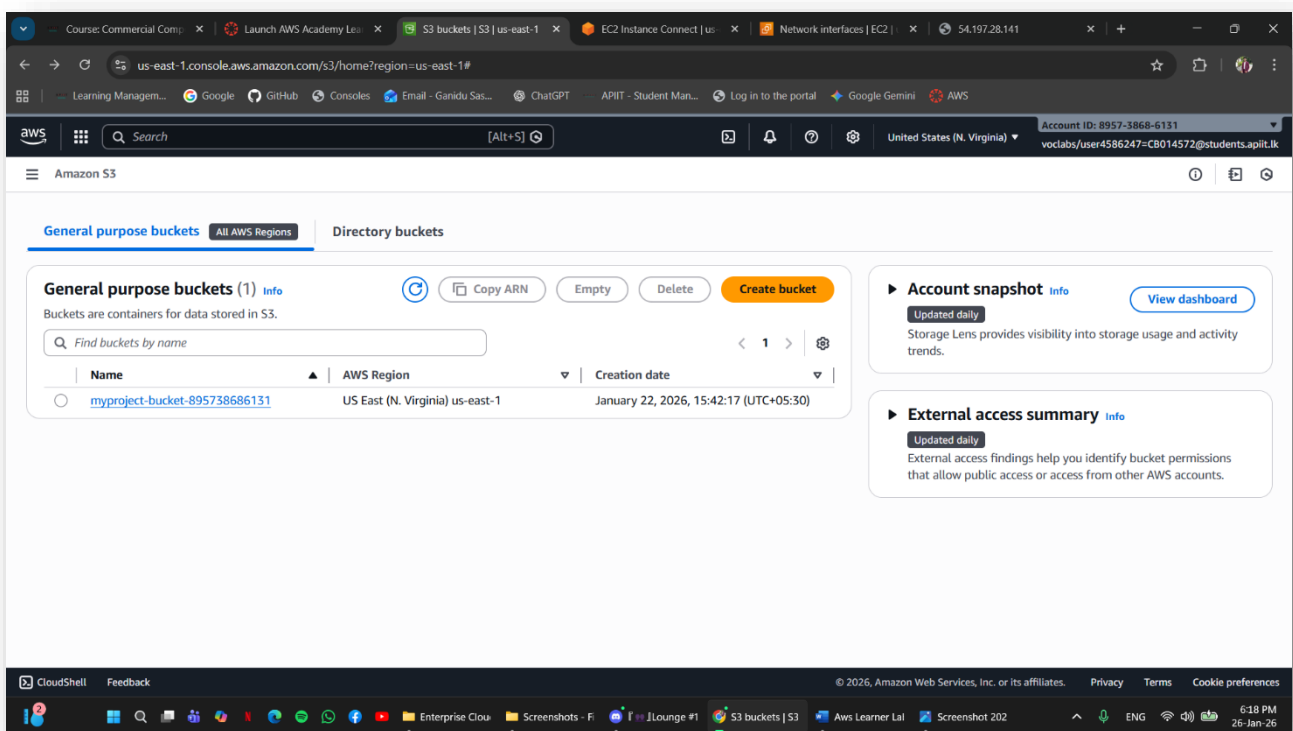


Figure 9 S3 Bucket Created

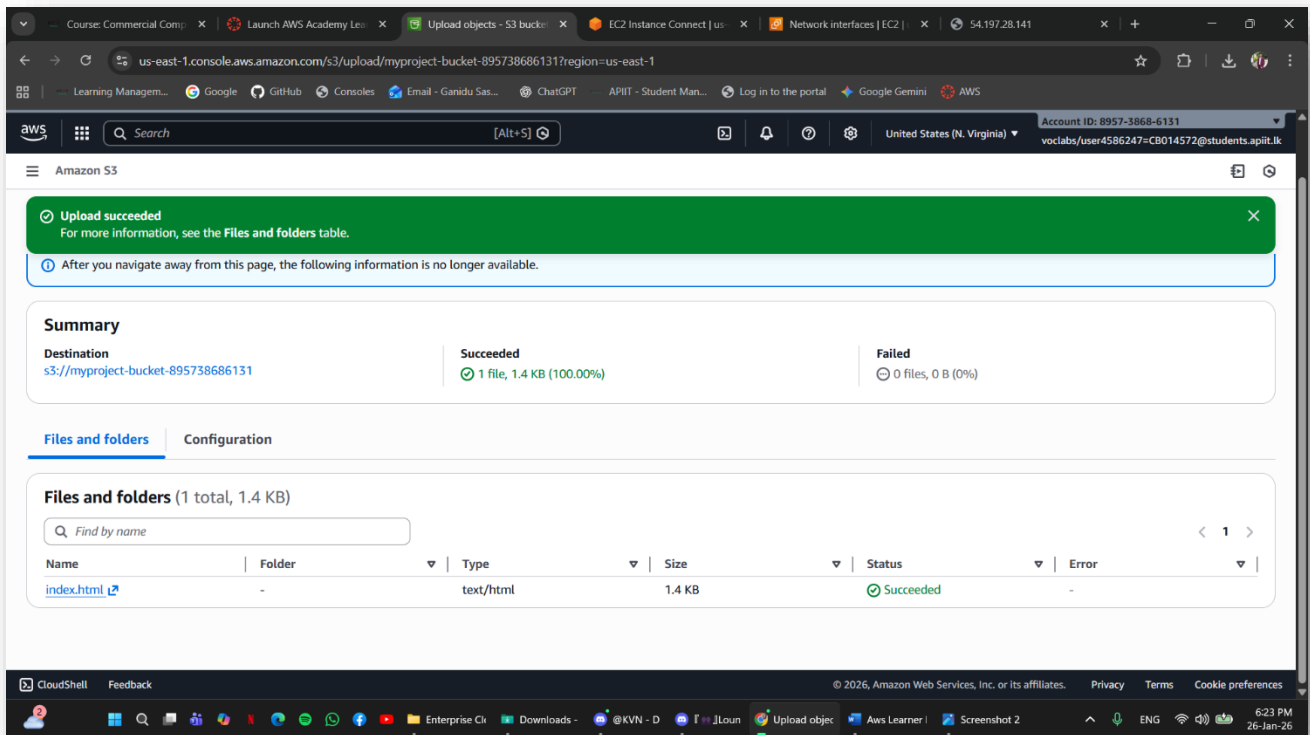


Figure 10 index.html Uploaded

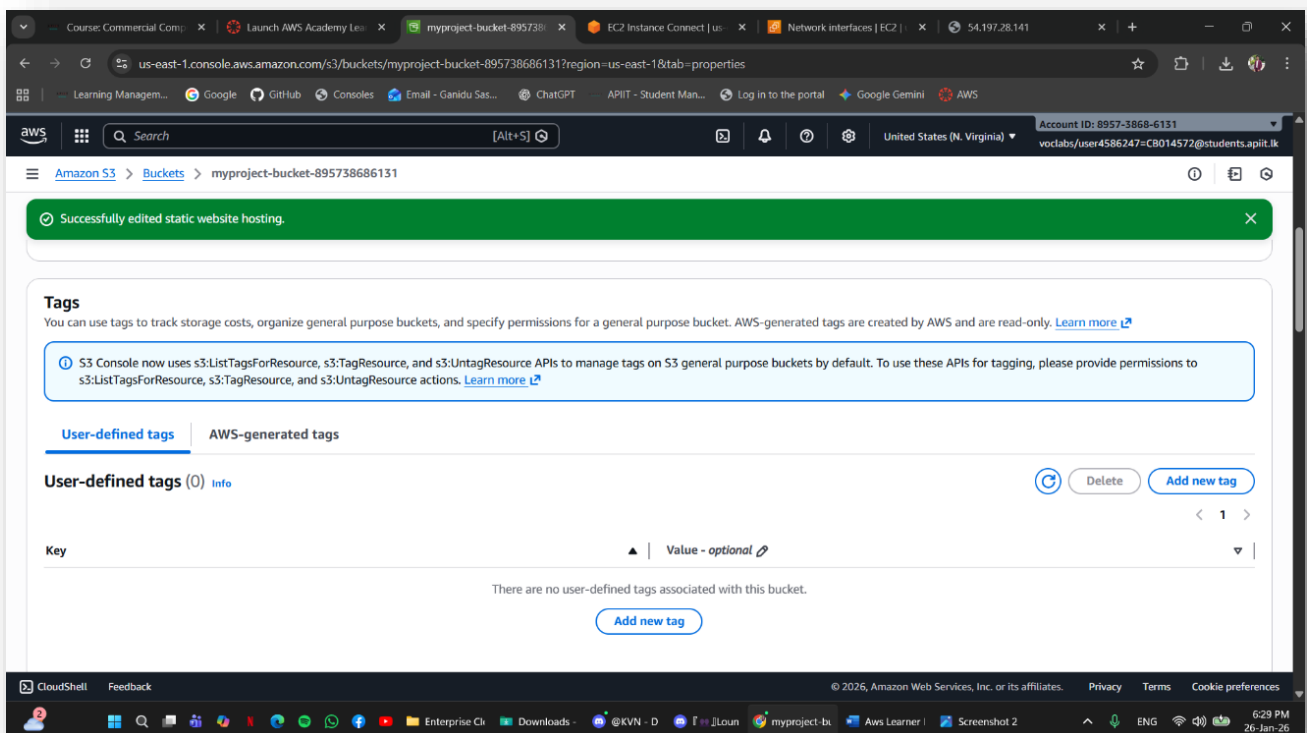


Figure 11 Static Website Hosting Enabled

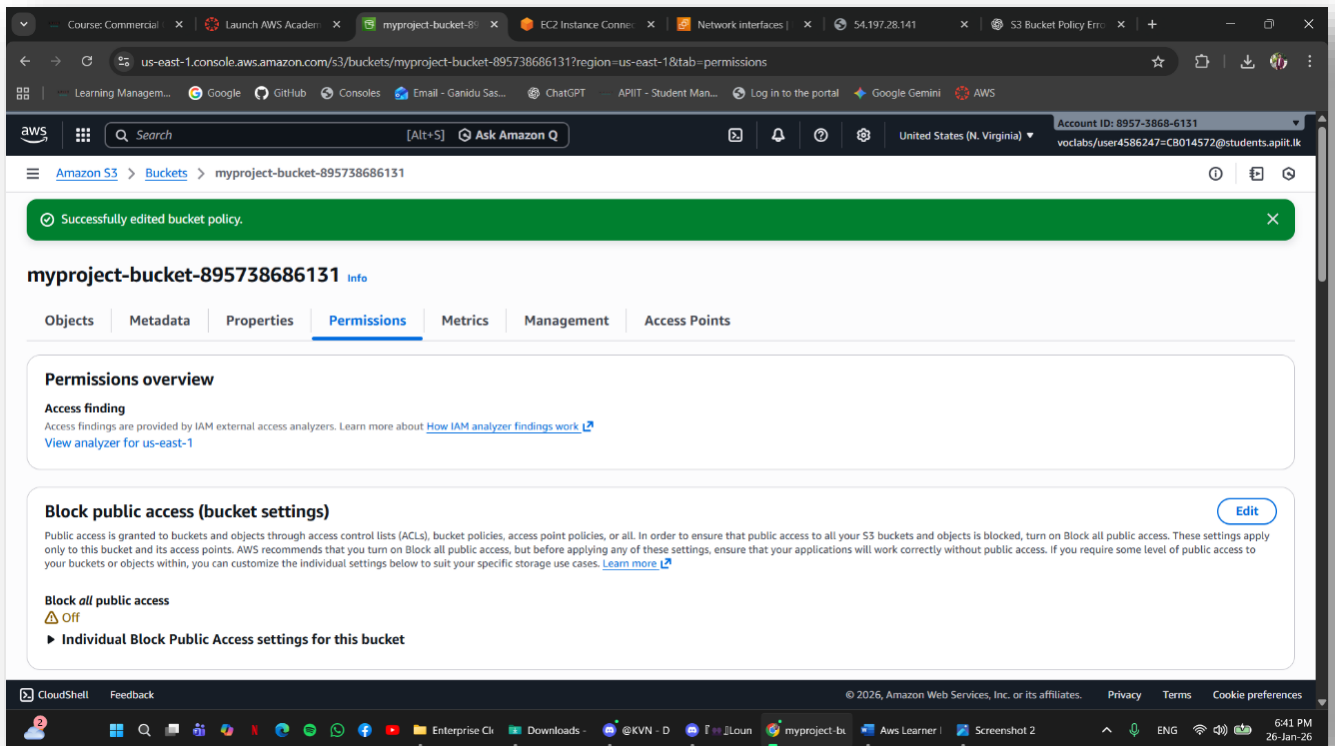


Figure 12 Bucket Policy JSON (ADD)

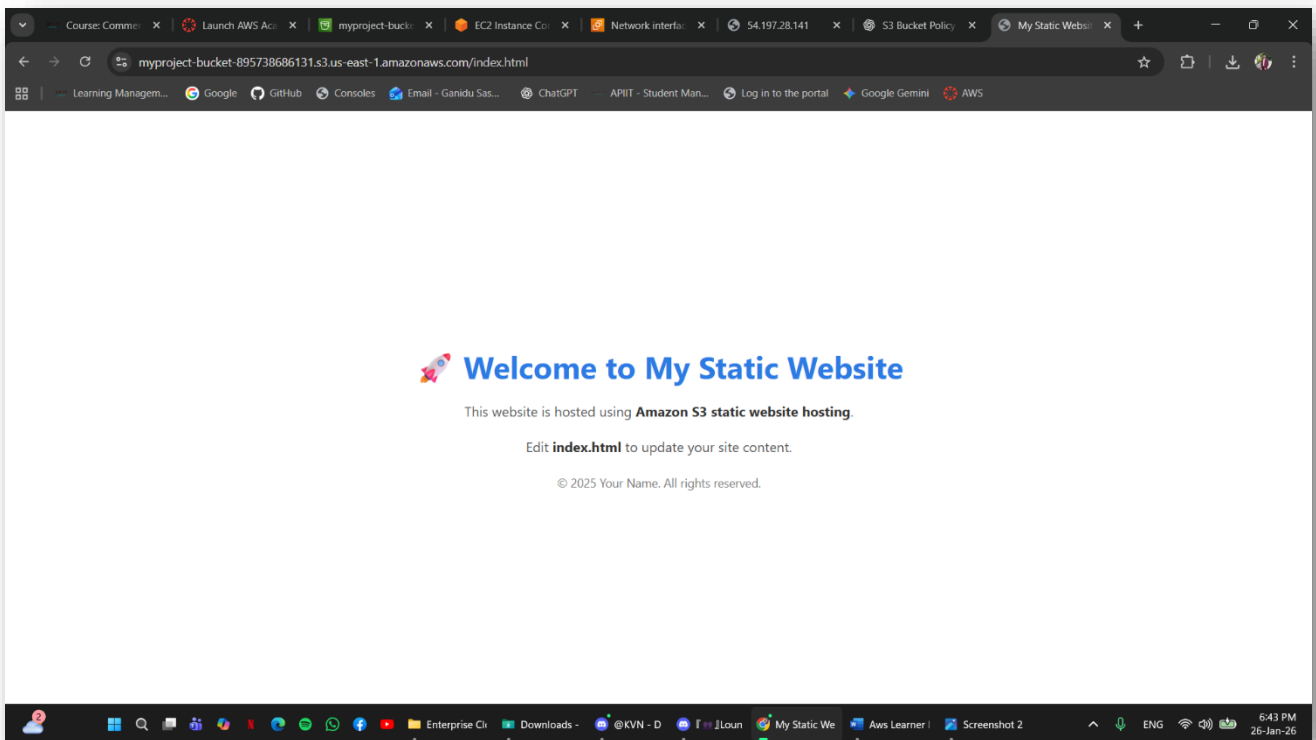


Figure 13 Browser showing S3 Website

7. Conclusion

The suggested cloud solution based on AWS offers a forward, looking plan that FreshMart can use for completely transforming its operations. The change is from a mainly manual, locally oriented business to a fully digital and robust enterprise. The infrastructure brings a scalable, secure, and cost, efficient platform that will uniquely help the business to continue growing in the face of a fierce market competition in Sri Lanka. This is done by giving the business the power to go global and be available for service 24/7.

Strategic Impact

- **Operational Efficiency:** Through adoption of AWS managed services such as Amazon RDS and S3, FreshMart effectively gets rid of the issue of hardware maintenance, thereby management can now focus their attention on core business growth and organic product innovation rather than server management.
- **Resilience and Trust:** The multi, layer design, supported by IAM and private subnets, makes sure that the most sensitive customer information is kept safe, thus resulting in customer loyalty over an extended period and meeting the requirements of the data protection laws that are likely to be enacted.
- **Performance Excellence:** By the use of CloudFront and Route 53, FreshMart is able to deliver high, performance services that have very low latency, hence the customers get a great shopping experience no matter where they are located.

Future Outlook

As the digital footprint of FreshMart grows, the base here is ready to change accordingly. The existing architecture is compatible with next, gen technologies like Machine Learning for targeted customer recommendations or IoT for live inventory tracking.

To sum up, this AWS cloud strategy is not only a solution for the infrastructure problems of today; it also sets up a future, ready environment that will keep FreshMart agile and competitive in a fast, evolving digital market.

8. Reference

Amazon, 2026. AWS. [Online]
Available at: <https://aws.amazon.com/what-is-cloud-computing/>
[Accessed 30 01 2026].

AWS, 2026. *Amazon EC2*. [Online]
Available at: <https://docs.aws.amazon.com/ec2/>
[Accessed 30 01 2026].

AWS, 2026. *Amazon Simple Storage Service Documentation*. [Online]
Available at: <https://docs.aws.amazon.com/s3/>
[Accessed 30 January 2026].

AWS, 2026. *AWS Security Documentation*. [Online]
Available at: <https://docs.aws.amazon.com/security/>
[Accessed 05 February 2026].

AWS, n.d. *Amazon Virtual Private Cloud Documentation*. [Online]
Available at: <https://docs.aws.amazon.com/vpc/>
[Accessed 05 February 2026].

AWS, n.d. *AWS Well-Architected Tool Documentation*. [Online]
Available at: <https://docs.aws.amazon.com/wellarchitected/>
[Accessed 05 February 2026].